

4th Grade Written Exam

2026 James Clemens Math Tournament

INSTRUCTIONS:

1. You will have 60 minutes to complete this examination.
2. This exam consists of 15 multiple choice questions and 3 tiebreaker questions. The multiple choice questions are each worth 4 points if answered correctly and -1 points if answered incorrectly. There is no penalty for unanswered questions. The tiebreaker questions are worth 0.1 points if answered correctly and 0 points if answered incorrectly or left unanswered.
3. Use a #2 pencil.
4. Calculators, cellular devices, books, notes, and other aids are prohibited during this exam. Scratch paper will be provided for calculations. If you require additional scratch paper, simply raise your hand and your proctor will assist you.
5. Mark your answers to the questions in the Scantron form provided. You may use the test booklets for scratch work, but only the answers marked in the Scantron form will be counted. Make sure to fill in the bubbles on the Scantron form **completely** and erase any stray marks.
6. Write your answers to the tiebreaker questions on the back of the Scantron form. Label the tiebreakers as TB1, TB2, and TB3.
7. NOTA denotes "None Of The Above."
8. Diagrams are not necessarily drawn to scale.
9. In the event of a tie, answers will be evaluated backwards starting from multiple choice question 15 to determine a winner.
10. Although this math tournament is intended to demonstrate your knowledge and skills in math, be sure to enjoy the experience and have fun!
11. You may keep the test booklet after the exam.

DO NOT TURN THIS PAGE UNTIL TOLD TO DO SO

1. Johnny Appleseed has an apple orchard with two kinds of trees: green apple and red apple. He planted 20 trees total. Each green apple tree produces 12 apples, and each red apple tree produces 9 apples. If Johnny harvested a total of 228 apples, how many red apple trees does he have?

(A) 3 (B) 4 (C) 5 (D) 6 (E) NOTA

2. Solve the expression:

$$\frac{3(83 + 2 \times 6 - 2^2)}{(10 - 3) \times 13}$$

(A) 1 (B) 2 (C) 3 (D) 4 (E) NOTA

3. There are 5 superheroes running in a race: Ant-Man, Black Panther, Captain America, Deadpool, and Echo. If there are no ties (every hero has the same chance of winning), what is the probability that Ant-Man ends up first and Black Panther ends up second?

(A) $\frac{1}{10}$ (B) $\frac{1}{30}$ (C) $\frac{1}{20}$ (D) $\frac{1}{5}$ (E) NOTA

4. Misty makes one Magic Muffin every fourth of an hour and recruits help from Danny, who makes one Magic Muffin every 30 minutes. If they both work together, how long does it take them to make one Magic Muffin?

(A) 5 mins (B) 8 mins (C) 10 mins (D) 12 mins (E) NOTA

5. The average of 12, 13, 7, 5, and n is 8.6. Find the value of n .

(A) 5 (B) 6 (C) 7 (D) 8 (E) NOTA

6. Pierson made a large pizza in his dorm and left it in the fridge untouched at 8:30am. However, the hungry kitchen ghost sneaks $\frac{1}{5}$ of his pizza the first hour, $\frac{1}{5}$ of the leftover pizza on the second hour, and so on. How much of the pizza will be leftover when Pierson checks on the pizza at 11:30am?

(A) $\frac{64}{125}$ (B) $\frac{81}{125}$ (C) $\frac{27}{125}$ (D) $\frac{125}{256}$ (E) NOTA

7. A triangle has two angles of 120° and 25° . What is the third angle?

(A) 30 (B) 35 (C) 45 (D) 55 (E) NOTA

8. How many times the digit 3 appear in the integers 1 – 100, inclusive?

(A) 10 (B) 11 (C) 15 (D) 20 (E) NOTA

9. Which of the following has the greatest value?

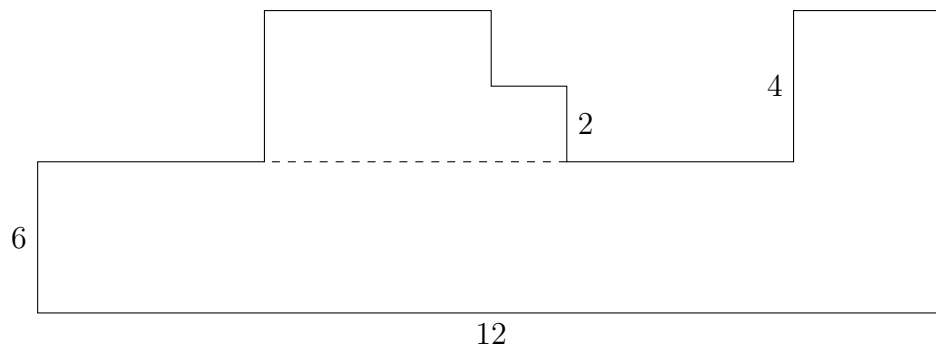
(A) 100×5 (B) 50×9 (C) 24×20 (D) 200×2

10. Snow White needs to choose her outfit for the ball! She has 5 ball gowns, 4 heels, 15 accessories, and 3 hairstyles to choose from. How many different outfits can she make?
(A) 850 (B) 900 (C) 950 (D) 980 (E) NOTA
11. Rudolph is on the run from the cops! Find the next number in this sequence to help him escape: 5, 3, 12, 9, 19, 15, 26, ?
(A) 29 (B) 30 (C) 35 (D) 21 (E) NOTA
12. The Three Little Pigs (Oinky, Pinky, and Squeaky) are building a new brick house to hide from the Big Bad Wolf. First, Oinky Pig brought 48 bricks, which is two-thirds of the total bricks they needed. Pinky Pig brought half of the rest of the bricks they needed. How many bricks did Squeaky Pig bring?
(A) 10 (B) 14 (C) 18 (D) 22 (E) NOTA
13. Time to test YOUR IQ! Can you evaluate: $999000 \times 11 + 1$?
(A) 1,098,901 (B) 10,989,001 (C) 10,999,001 (D) 10,999,000
(E) NOTA
14. Rumi has a magic number that Zoey has to guess. She says, "I am thinking of a number. If you multiply my number by 3, then add 12, you get 75. What's the number I'm thinking of?" What number should Zoey correctly guess?
(A) 18 (B) 21 (C) 24 (D) 27 (E) NOTA
15. Sophia the First has a mango tree in front of her window. She watches 40 bluebirds land on one branch. Then, 10 fly away, 8 come back to the tree, 15 more arrive, and 7 fly away. How many bluebirds are on the tree now?
(A) 46 (B) 48 (C) 50 (D) 54 (E) NOTA

Tiebreakers:

- TB1. How many muffins can Danny and Misty make in 3 hours? (References problem 4 on this test)

TB2. Find the total perimeter of the figure (all sides meet at right angles):



TB3. What is $1.1 \times 1.1 \times 100$?